

xPrime Number Identifier Circuit

PC/CP220 Lab 5

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Description:

Finding prime numbers is a common task in mathematics. This circuit will identify the prime numbers between zero and fifteen. A number is defined as prime if it has exactly two divisors, itself and one.

Prime/Composite/None:

Number	Binary(a_3, a_2, a_1, a_0) Prime	P/C/N(Prime/Composite, None)
0	0000	N
1	0001	N
2	0010	P
3	0011	P
4	0100	C
5	0101	P
6	0110	C
7	0111	P
8	1000	C
9	1001	C
10	1010	C
11	1011	P

12	1100	C
13	1101	P
14	1110	C
15	1111	C

Inputs:

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t1:not(a3) and not(a2) and a1;
t2:not(a3) and a2 and a0;
t3:not(a2) and a1 and a0;
t4:a2 and not(a1) and a0;

prime:t1 or t2 or t3 or t4;

prime,a0=false,a1=false,a2=false,a3=false;

prime,a0=true,a1=false,a2=false,a3=false;
prime,a0=false,a1=true,a2=false,a3=false;
prime,a0=true,a1=true,a2=false,a3=false;
prime,a0=false,a1=false,a2=true,a3=false;
prime,a0=true,a1=false,a2=true,a3=false;

prime,a0=false,a1=true,a2=true,a3=false;
prime,a0=true,a1=true,a2=true,a3=false;
prime,a0=false,a1=false,a2=false,a3=true;
prime,a0=true,a1=false,a2=false,a3=true;
prime,a0=false,a1=true,a2=false,a3=true;

prime,a0=true,a1=true,a2=false,a3=true;
prime,a0=false,a1=false,a2=true,a3=true;
prime,a0=true,a1=false,a2=true,a3=true;
prime,a0=false,a1=true,a2=true,a3=true;
prime,a0=true,a1=true,a2=true,a3=true;

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Output:

$\neg a_3 \wedge \neg a_2 \wedge a_1$	
$\neg a_3 \wedge a_2 \wedge a_0$	
$\neg a_2 \wedge a_1 \wedge a_0$	
$a_2 \wedge \neg a_1 \wedge a_0$	
$\neg a_3 \wedge \neg a_2 \wedge a_1 \vee \neg a_3 \wedge a_2 \wedge a_0 \vee \neg a_2 \wedge a_1 \wedge a_0 \vee a_2 \wedge \neg a_1 \wedge a_0$	
false	
false	
true	
true	
false	
true	
false	
true	
false	
false	
false	
true	
false	
true	
false	
false	

Output explained:

Output 1: $a_0 = \text{False}$, $a_1 = \text{False}$, $a_2 = \text{False}$, $a_3 = \text{False}$

- The above boolean algebra is equivalent to 0000 in binary which is equal to the decimal value of 0. Which is neither a composite nor a prime number which is why the output came back as False.

Output 2: a0 = True, a1 = False, a2 = False, a3 = False

- The above boolean algebra is equivalent to 0001 in binary which is equal to the decimal number 1. Which is neither composite nor a prime number, that's why the output came back as False

Output 3: a0 = false, a1 = true, a2 = false, a3 = false;

- The above boolean algebra is equivalent to 0010 in binary which is equal to the decimal number 2. Which is Prime, that's why the output was True.

Output 4: a0 = true, a1 = true, a2 = false, a3 = false;

- The above boolean algebra is equivalent to 0011 in binary which is equal to the decimal number 3. Which is Prime, that's why the output was True.

Output 5: a0 = false, a1 = false, a2 = true, a3 = false;

- The above boolean algebra is equivalent to 0100 in binary which is equal to the decimal number 4. Which is Composite, that's why the output was False.

Output 6: a0 = true, a1 = false, a2 = true, a3 = false;

- The above boolean algebra is equivalent to 0101 in binary which is equal to the decimal number 5. Which is Prime, that's why the output was True.

Output 7: a0 = false, a1 = true, a2 = true, a3 = false;

- The above boolean algebra is equivalent to 0110 in binary which is equal to the decimal number 6. Which is Composite, that's why the output was False.

Output 8: a0 = true, a1 = true, a2 = true, a3 = false;

- The above boolean algebra is equivalent to 0111 in binary which is equal to the decimal number 7. Which is Prime, that's why the output was True.

Output 9: a0 = false, a1 = false, a2 = false, a3 = true;

- The above boolean algebra is equivalent to 1000 in binary which is equal to the decimal number 8. Which is Composite, that's why the output was False.

Output 10: a0 = true, a1 = false, a2 = false, a3 = true;

- The above boolean algebra is equivalent to 1001 in binary which is equal to the decimal number 9. Which is Composite, that's why the output was False.

Output 11: a0 = false, a1 = true, a2 = false, a3 = true;

- The above boolean algebra is equivalent to 1010 in binary which is equal to the decimal number 10. Which is Composite, that's why the output was False.

Output 12: a0 = true, a1 = true, a2 = false, a3 = true;

- The above boolean algebra is equivalent to 1011 in binary which is equal to the decimal number 11. Which is Prime, that's why the output was True.

Output 13: a0 = false, a1 = false, a2 = true, a3 = true;

- The above boolean algebra is equivalent to 1100 in binary which is equal to the decimal number 12. Which is Composite, that's why the output was False.

Output 14: a0 = true, a1 = false, a2 = true, a3 = true;

- The above boolean algebra is equivalent to 1101 in binary which is equal to the decimal number 13. Which is Prime, that's why the output was True.

Output 15: a0 = false, a1 = true, a2 = true, a3 = true;

- The above boolean algebra is equivalent to 1110 in binary which is equal to the decimal number 14. Which is Composite, that's why the output was False.

Output 16: a0 = true, a1 = true, a2 = true, a3 = true;

- The above boolean algebra is equivalent to 1111 in binary which is equal to the decimal number 15. Which is Composite, that's why the output was False.

Conclusion:

The Prime Number Identifier Circuit successfully identifies prime, composite, and inapplicable numbers out of the range from 0 to 15. This simplification work is correct as it returns the value “True” for all prime numbers like 2, 3, 5, 7, 11 and 13; the values with exact two divisors that is 1 and the number itself while the common numbers having more than two divisors are returned “False”. Also, if the input for the circuit is 0 or 1 it will correctly read it as not a prime number and not a composite number by returning a “False” for both. This project is evidence of how the circuit works in number classification within a specific range; further still, the above examples go to show how digital circuits can be used to solve mathematical problems such as identification of prime numbers.